Week 7 · Day 5 Homework

The other two cloud AI platforms for UC1, ADR-002, and the pilot teardown

Kestrelmoor Bank engagement · Phase 2 · prepared 10 November 2026 by Mithran Sadasivam

What is in this document

- **Part 1 - Platform comparison** (docs/eval/PLATFORM_COMPARISON.md): Microsoft Foundry and Google's Gemini Enterprise Agent Platform against the bank's eight rules and the monthly model cost. Every cell has a source and the date checked.
- **Part 2 - ADR-002** (docs/adr/ADR-002-pilot-platform.md): the decision record with the new section "Other platforms considered". The Decision is unchanged; the Decision record checker gives 9 of 9.
- **Part 3 - Teardown record** (docs/eval/TEARDOWN_RECORD.md): what I could and could not prove. The four proof commands could not be run as written; the real error output is recorded and nothing is invented.

Headline results

- **Both other platforms fail rule 1** (Claude processed in the UK only). Foundry offers Claude as Global Standard or US Data Zone only; Google offers Sonnet 5 on global or US/EU multi-region only, and the EU multi-region excludes the UK.
- **Bedrock stays.** The model costs about the same everywhere (about £51 on a global endpoint, £56 with a residency option) so cost does not change the answer.
- **The teardown is not proved.** Terraform and the AWS CLI are not installed here, no Terraform state exists in the repository, and the lab account's service control policy denies the read calls. The record says exactly that and lists the four commands to run.

The other two platforms, for UC1

Checked on: 10 November 2026 · **By:** Mithran Sadasivam (with Claude Code doing the page reading)

Every fact below carries a source tag like [S3]; the sources are listed at the end with the date I opened them. Microsoft Foundry was Azure AI Foundry; Google's Gemini Enterprise Agent Platform was Vertex AI. Most articles still use the old names.

The question

Could the UC1 pilot run on Microsoft Foundry or on Google's Gemini Enterprise Agent Platform instead of Amazon Bedrock, under the bank's eight rules?

Against the bank's rules

The first rule is about the **model's** endpoint, not the platform's map. Both platforms have UK Regions, but not for Claude.

Rule	Amazon Bedrock (today)	Microsoft Foundry	Google Gemini Enterprise Agent Platform
A Claude model processed in the UK only	Yes: Claude Sonnet 5, in-Region in London [S19]	**No.** Claude Sonnet 5 (Hosted on Azure) deploys as Global Standard, where "Microsoft processes prompts and responses in any Azure region where you deploy the model", or Data Zone Standard (US) only. The only Europe Region ticked for Claude is Sweden Central; UK South and UK West are in the table but not ticked for any Claude model. The Europe Data Zone column says "Not available". The Anthropic-hosted version "might be processed outside Azure" [S1] [S2] [S3]	**No.** Sonnet 5 is listed for the US multi-region, the Europe multi-region and the global endpoint only [S11]. Regional endpoints serve "Claude Sonnet 4.6 and earlier models only" [S9]. Google's residency page lists London for no Claude model, and says the EU multi-region excludes the UK ("Geographies outside the European Union political boundary, including the United Kingdom and Switzerland, are excluded") [S10]
Private network access to the model	Yes: VPC interface endpoint [S19]	Partly confirmed: a private endpoint (Private Link) can be added to the Foundry account and public access disabled [S7]. The page does not say whether it covers Marketplace-billed Claude deployments, so for Claude this is **not confirmed**	**Not confirmed.** The Private Service Connect page for Agent Platform does not mention Claude, rawPredict or partner models, and says PSC backends are not supported with online prediction endpoints [S17]
An IAM-style role, no long-lived keys	Yes: Lambda execution role [S19]	Yes: Microsoft Entra ID (keyless) is supported, with the Cognitive Services User role; API keys are the alternative [S1]	Yes: an IAM role with `aiplatform.endpoints.predict` (Agent Platform User) for the calling identity, including a service account [S9] [S15]
Secrets in a managed vault	Yes: Secrets Manager [S19]	Yes: Azure Key Vault stores "tokens, passwords, certificates, API keys, and other secrets", access via Entra ID [S8]	Yes: Secret Manager stores "API keys, usernames, passwords, certificates", access by IAM [S16]
Every model call logged	Yes: model invocation logging [S19]	**Not confirmed.** The page I wanted on diagnostic logging returned 404. Cost Management gives per-model token and cost breakdowns, but that is billing, not a call log [S5] [S6]	Yes: request-response logging supports Anthropic Claude through `rawPredict`, saved to BigQuery; for Claude it is set up by REST only. It "isn't available with VPC Service Controls" [S14] [S18-vpcsc]
Guardrails (prompt attacks, PII masking)	Yes: Bedrock Guardrails [S19]	**No built-in filter for Claude.** Microsoft says "Foundry doesn't provide built-in content filtering for Claude models at deployment time" and tells you to configure content safety yourself; Claude uses Anthropic's safety systems [S4]. PII masking not confirmed	Partly: Model Armor detects prompt injection and jailbreaks and can redact PII [S13]. That page does not say it works with Claude on Agent Platform, so for Claude this is **not confirmed**
A spending limit that stops calls	Yes: budget action [S19]	**No.** Budgets only alert; Microsoft says there is no hard limit and a stop needs custom automation from action groups [S6]	**Yes, per its docs:** a spend cap on `aiplatform.googleapis.com` pauses all new usage of the service in that project. It is not instant, is based on estimated list-price cost, and overages are still billed [S12]. The page does not say whether it is generally available
Terraform support	Yes: hashicorp/aws [S19]	Yes: Microsoft documents deploying Claude on Foundry with Bicep or Terraform [S1]	Not checked: the Terraform Registry page would not load, so I cannot give a version. The Google provider is widely used, but I have no source from this session

Cost for UC1

Volume as on Day 5: 8,600 documents a month, 1,680 tokens in and 420 out each = 14.448 million input and 3.612 million output tokens. Exchange rate 0.78 GBP per USD, the course's illustrative rate [S19].

	Amazon Bedrock	Microsoft Foundry	Google Gemini Enterprise Agent Platform
Model and price (USD per million tokens, in / out)	Claude Sonnet 5 in-Region (London): 2.20 / 11.00 [S19]	Claude Sonnet 5: list 2.00 / 10.00 on Global Standard; US Data Zone Standard is 1.1x = 2.20 / 11.00 [S18] [S5]	Claude Sonnet 5: list 2.00 / 10.00 on the global endpoint; multi-region "us" or "eu" costs 10 percent more = 2.20 / 11.00 [S9] [S18]

	Amazon Bedrock	Microsoft Foundry	Google Gemini Enterprise Agent Platform
Model cost a month at 8,600 documents (GBP)	£56 (USD 71.52)	Global: £51 (USD 65.02), **but not UK**. US Data Zone: £56, **but US only**. Neither is usable under rule 1	Global: £51 (USD 65.02), **but not UK**. EU multi-region: £56 (USD 71.52), **but excludes the UK**. Neither is usable under rule 1

How the numbers are made: 14.448 x 2.00 + 3.612 x 10.00 = USD 65.02, and 14.448 x 2.20 + 3.612 x 11.00 = USD 71.52; times 0.78 gives £50.71 and £55.79.

Price caveats:

- Sonnet 5 launched with introductory pricing of 2 / 10 "through August 31, 2026". Anthropic now says that is "the standard price" and the planned rise to 3 / 15 on 1 September 2026 "will not occur" [S18]. So 2 / 10 is today's list price, with no end date.
- I did not open Google's own pricing page or Microsoft's Marketplace CCU price page. Both clouds bill at Anthropic's published per-model rates [S5] [S18], so I used the rates from Anthropic's page. A real quote from each cloud is still to be asked for. On Foundry, CCU is a billing unit at USD 0.01 per CCU, not a price change [S5] [S18].
- Platform extras (networking, logging, a guardrail service) are not in these numbers on any platform.

Verdict

Neither platform can replace Bedrock for UC1 today. Both fail the first rule: there is no way to have Claude Sonnet 5 processed only in the UK on Foundry (global or US Data Zone only) or on Google (global or US/EU multi-region only; the EU one excludes the UK). Both could probably meet most of the other rules with extra work, but three are unconfirmed for Claude (private access on both, call logging on Foundry, Model Armor with Claude on Google), Foundry has no built-in Claude guardrail and no spending stop, and the price difference is zero once a residency option is chosen. What would have to change first: Anthropic and Microsoft, or Google, must offer Claude in a UK-only deployment.

Sources

Every page below was opened on 10 November 2026 unless it says otherwise.

- [S1] Deploy and use Claude models in Microsoft Foundry, Microsoft Learn (page dated 2026-09-21). https://learn.microsoft.com/en-us/azure/foundry/foundry-models/how-to/use-foundry-models-claude
- [S2] Compare hosting options for Claude models in Microsoft Foundry, Microsoft Learn. https://learn.microsoft.com/en-us/azure/foundry/foundry-models/concepts/claude-models-hosting-comparison
- [S3] Foundry Models from partners and community, "Region availability by deployment type", Microsoft Learn. https://learn.microsoft.com/en-us/azure/foundry/foundry-models/concepts/models-from-partners
- [S4] Claude models in Microsoft Foundry ("Responsible AI considerations"), Microsoft Learn. https://learn.microsoft.com/en-us/azure/foundry/foundry-models/concepts/claude-models
- [S5] Claude consumption units (CCU) billing in Microsoft Foundry, Microsoft Learn (updated 2026-11-09). https://learn.microsoft.com/en-us/azure/foundry/foundry-models/concepts/claude-models-billing
- [S6] Plan and manage costs, Microsoft Foundry, Microsoft Learn ("Create budgets"). https://learn.microsoft.com/en-us/azure/foundry/concepts/manage-costs
- [S7] How to configure network isolation for Microsoft Foundry, Microsoft Learn. https://learn.microsoft.com/en-us/azure/foundry/how-to/configure-private-link
- [S8] Azure Key Vault overview, Microsoft Learn. https://learn.microsoft.com/en-us/azure/key-vault/general/overview
- [S9] Claude Models on Google Cloud (Google managed), Claude Platform docs. https://platform.claude.com/docs/en/build-with-claude/claude-on-vertex-ai
- [S10] Data residency, Google Cloud. https://docs.cloud.google.com/vertex-ai/generative-ai/docs/learn/data-residency
- [S11] Anthropic Claude models on Agent Platform, Claude Sonnet 5 model card (locations), Google Cloud. https://docs.cloud.google.com/gemini-enterprise-agent-platform/models/partner-models/claude/sonnet-5
- [S12] Manage spend cap budgets, Google Cloud Billing. https://docs.cloud.google.com/billing/docs/how-to/budgets-spend-caps
- [S13] Model Armor overview, Google Cloud. https://docs.cloud.google.com/model-armor/overview
- [S14] Request-response logging, Google Cloud. https://docs.cloud.google.com/vertex-ai/generative-ai/docs/multimodal/request-response-logging
- [S15] Use partner models (Claude) on Agent Platform, Google Cloud. https://docs.cloud.google.com/gemini-enterprise-agent-platform/models/partner-models/use-partner-models
- [S16] Secret Manager overview, Google Cloud. https://docs.cloud.google.com/secret-manager/docs/overview

- [S17] Private Service Connect for Agent Platform endpoints, Google Cloud. https://docs.cloud.google.com/vertex-ai/docs/general/psc-endpoints
- [S18] Pricing, Claude Platform docs (model table, footnote 3, cloud-platform pricing). https://platform.claude.com/docs/en/about-claude/pricing
- [S18-vpcsc] VPC Service Controls with Agent Platform, Google Cloud ("Request and response logging isn't available with VPC Service Controls"). https://docs.cloud.google.com/vertex-ai/docs/general/vpc-service-controls
- [S19] Course files: data/Day05_Slide10_bedrock_models.csv (Bedrock model card and price, checked 5 Nov 2026) and data/Day02_Slide21_price_sheet.csv (exchange rate 0.78).

What I could not confirm

- Foundry: call logging for Claude deployments (page returned 404); PII masking; whether the Private Link setup covers Marketplace Claude deployments.
- Google: Private Service Connect for Claude; Model Armor with Claude; the Terraform provider version; whether spend caps are generally available; Google's own price page.
- Both platforms change monthly: re-check on the date in ADR-002.

ADR-002: Run the UC1 pilot on Amazon Bedrock with Claude Sonnet 5, in-Region in London

- **Status:** proposed
- **Date:** Day 5 of the engagement; the section "Other platforms considered" was added on 10 November 2026
- **Deciders:** Helen Achebe (MLRO) for the rules, Martin Hale (COO) for the cost
- **Supersedes in part:** ADR-001 (the model choice)

Context

ADR-001 chose Textract plus Claude Haiku. On Day 5 the pilot went onto the bank's cloud, AWS London (eu-west-2), and the Bedrock model cards show that Claude Haiku 5.5 is not available in-Region in London: it only runs through the EU profile, which routes to seven EU Regions. Helen's rule is that customer documents must not leave the UK, so the model must be called in-Region in London. The pilot has to prove that UC1 can read a proof of address under all eight pilot rules (UK processing, private access, least privilege, no plain-text secrets, every call logged, guardrails, a spending limit, and a person decides), at a budget of USD 20 a month agreed with Martin.

Options considered

	Option A · Claude Haiku 5.5 (ADR-001's model)	Option B · Claude Sonnet 5	Option C · Claude Opus 5.5
Model ID called from London	eu.anthropic.claude-haiku-5-5 (EU profile)	anthropic.claude-sonnet-5 (in-Region)	anthropic.claude-opus-5-5 (in-Region)
Where the call is processed	Any of 7 EU Regions, not only the UK	London only	London only
Meets the client rules?	No: breaks the UK-only rule	Yes	Yes
Model cost a month (GBP, 8,600 documents)	£3 (USD 0.11 / 0.55 per million tokens)	£56 (USD 2.20 / 11.00)	£112 (USD 4.40 / 22.00, list 4 / 20 plus 10 percent)
Main risk	Not allowed for customer documents	Costs about 20 times Option A	Twice the cost of Option B for a gain nobody has measured

Costs use 8,600 documents a month at 1,680 input and 420 output tokens each, and the course exchange rate of 0.78 GBP per USD. Sources: data/Day05_Slide10_bedrock_models.csv (Bedrock model cards, checked 5 Nov 2026) and the Anthropic pricing page for Opus 5.5 (checked 10 Nov 2026).

Decision

Option B, Claude Sonnet 5 called in-Region in London on Amazon Bedrock. Three reasons:

1. **Rules:** it is the cheapest option that keeps processing in the UK. Option A fails the UK-only rule.
2. **Quality:** the Day 4 evaluation (prompt extract_v2, 98.5 percent of fields correct on 100 cases) was not run on this model. Before the pilot reads real documents, run `python -m scripts.run_eval --prompt extract_v2` on Sonnet 5 and accept it only if it scores at least as well. Until then this reason is a plan, not a result.
3. **Cost:** £56 a month for the model at the full 8,600 documents (USD 71.52). The pilot week reads far fewer documents and stays under the USD 20 budget; Martin must agree the full-volume figure before production.

Also decided for the pilot:

- **Guardrail:** policy uc1_v1 (infra/guardrails/uc1_v1.json), read by both Terraform and the scripted guardrail. Its probe results (`python -m scripts.probe_guardrail uc1_v1`) and its effect on the evaluation are not yet recorded here, so they must be run and written down before go-live.
- **Knowledge base:** none. UC1 reads one document at a time and compares it with the application; there is no collection of documents to search, so a retrieval layer would add cost and a place for customer data to live without adding anything the model needs.
- **Spend:** a monthly budget of USD 20 (variable budget_usd). An email alert at 80 percent of actual spend and at 100 percent of forecast spend. At 100 percent of actual spend the budget action attaches a policy that denies the pilot's role every Bedrock call. Billing data reaches AWS a few times a day, so this is a backstop, not a real-time cap.

Other platforms considered

Facts are taken from docs/eval/PLATFORM_COMPARISON.md (checked 10 November 2026, each cell has a source there). Correct a fact in that file, not here.

	Amazon Bedrock	Microsoft Foundry	Google Gemini Enterprise Agent Platform
Claude processed in the UK only	Yes: Sonnet 5 in-Region in London	No: Global Standard (any Azure Region) or US Data Zone only; Europe Data Zone not available; no UK Region lists Claude	No: global, US or EU multi-region only; the EU multi-region excludes the UK; regional endpoints stop at Sonnet 4.6
Private access to the model	Yes: VPC interface endpoint	Private Link exists for the Foundry account; not confirmed for Claude	Private Service Connect: not confirmed for Claude
Role, no long-lived keys	Yes: Lambda execution role	Yes: Entra ID	Yes: IAM role and service account
Secrets in a vault	Yes: Secrets Manager	Yes: Key Vault	Yes: Secret Manager
Every model call logged	Yes: invocation logging	Not confirmed	Yes: request-response logging to BigQuery
Guardrails	Yes: Bedrock Guardrails	No built-in filter for Claude	Model Armor exists; with Claude not confirmed
Spending limit that stops calls	Yes: budget action	No: alerts only	Yes: spend cap, not instant
Terraform	Yes: hashicorp/aws	Yes: Bicep or Terraform documented	Not checked
Model cost a month (8,600 documents)	£56	£51 on the global endpoint, £56 on the US Data Zone: neither usable	£51 on the global endpoint, £56 on the EU multi-region: neither usable

Verdict. Bedrock stays: both other platforms fail the UK-only rule for Claude, so the Decision above is unchanged.

What would reopen it. Any one of these, each of which Helen can check on a public page:

1. Microsoft's "Region availability by deployment type" table lists Claude Sonnet 5 (or the model we run) as deployable in UK South or UK West, or adds a UK or EU Data Zone for Claude whose processing stays in the UK.
2. Google's data residency page or the Claude Sonnet 5 model card lists London (europe-west2), or a UK-only endpoint, for the model we run.
3. Bedrock stops offering our model in-Region in London.

When I will look again. On 10 December 2026, and again at the pilot review before any real customer document is used. These platforms change every month, so the check is by date, not by event.

Consequences

- **Good:** the model is processed only in London; every rule in the pilot is met by one provider and one Terraform provider; the budget stops calls itself; the decision shows it was compared with the other two clouds and says what would change it.
- **Bad:** Option B costs about 20 times Haiku; the Day 4 quality figures and the guardrail probe results still have to be produced for this model.
- **To watch:** the three reopen triggers above, on 10 December 2026; the cost if volume rises to the January peak of 17,200 documents (about £112 a month for the model); and the open points in PLATFORM_COMPARISON.md that I could not confirm.

Pilot teardown record

Date and time: 10 November 2026, about 13:55 (machine local time) · **Account:** ending 4332 · **Region:** eu-west-2

Honest summary: I could not run the four proof commands as written, and I have not pasted any output I did not get. Terraform and the AWS CLI are not installed on this machine, there is no Terraform state anywhere in the repository, and the AWS account's organisation policy denies every read call I tried. The evidence below is what I could really observe. The checks must be run by the student in the lab account (or by the trainer) to complete this record.

1. Destroy

- Command: `terraform destroy` in infra/ — **not run.**
- Resources destroyed (the last line of the output): **none, because nothing was applied from this machine.** Evidence:
  - `terraform` is not on the PATH (`terraform: command not found`).
  - No `terraform.tfstate`, `.terraform/`, `terraform.tfvars`, `pilot.tfplan` or `pilot.plan.json` exists anywhere under the project. Only the `.tf` files and `terraform.tfvars.example` are in infra/.
  - One trace of Terraform use does exist: `infra/.terraform.lock.hcl`, which `terraform init` creates. So Terraform was initialised in infra/ at some point, but it left no `.terraform/` folder and no state, which is consistent with no apply (or with files removed afterwards; I cannot tell which).
  - The Lambda package (`infra/build/lambda.zip`) was never built: infra/build does not exist.
  - Git history for this project stops at "Day 5 start"; no commit applies the pilot.
  - Rule 8 in CLAUDE.md says `terraform apply` and `terraform destroy` are never run without showing the plan and asking, and I was not asked to apply or destroy, so I did not install Terraform to do it.

2. Prove nothing is left

Check	Command	Result
Terraform state is empty	`terraform state list`	**Not run** (Terraform not installed). There is no state file at all, which is what an empty state means here.
No VPC endpoints with the pilot's tag	`aws ec2 describe-vpc-endpoints --region eu-west-2 --filters Name=tag:project,Values=fde-uc1-pilot --query "VpcEndpoints[].VpcEndpointId"`	**Could not be checked.** The AWS CLI is not installed, so I made the same call with boto3 (`describe_vpc_endpoints`, same tag filter, eu-west-2). Real output: `UnauthorizedOperation - You are not authorized to perform this operation. User: arn:aws:iam::...4332:user/fde-student is not authorized to perform: ec2:DescribeVpcEndpoints with an explicit deny in a service control policy`
No pilot function	`aws lambda get-function --function-name uc1-pilot --region eu-west-2`	**Could not be checked.** boto3 `get_function` for uc1-pilot returned: `AccessDeniedException - User: arn:aws:iam::...4332:user/fde-student is not authorized to perform: lambda:GetFunction on resource: arn:aws:lambda:eu-west-2:...4332:function:uc1-pilot with an explicit deny in a service control policy`. This is not `ResourceNotFoundException`, so it proves nothing about whether the function exists.
Invocation logging is off	`aws bedrock get-model-invocation-logging-configuration --region eu-west-2`	**Could not be checked.** boto3 returned: `AccessDeniedException - ... is not authorized to perform: bedrock:GetModelInvocationLoggingConfiguration ... with an explicit deny in a service control policy`

I also tried `ec2:DescribeVpcs` (tag project=fde-uc1-pilot) and `lambda:ListFunctions`; both were denied by the same service control policy. That policy is set at the organisation level, so it looks like the course lab account has been locked after class; I cannot change it and did not try to.

What this means: **I have not proved the pilot is gone, and I have not proved it is still there.** All I can show is that nothing was applied from this machine. If someone else applied the pilot from another machine or session, its resources could still exist, and the two private endpoints would keep costing about USD 0.48 a day.

To finish the record, run these from a shell that has Terraform and an AWS identity allowed to read (not the locked student user), and paste the real output over the "Could not be checked" cells:

```
cd infra
terraform state list
aws ec2 describe-vpc-endpoints --region eu-west-2 --filters Name=tag:project,Values=fde-uc1-pilot --query "VpcEndpoints[]
aws lambda get-function --function-name uc1-pilot --region eu-west-2
aws bedrock get-model-invocation-logging-configuration --region eu-west-2
```

Proved when: no output; `[]`; `ResourceNotFoundException`; no `loggingConfig`. The last check matters because invocation logging is an account-and-Region setting: if it were left on, every Bedrock call in the account would keep writing logs.

3. What the day cost

- From Billing and Cost Management -> Bills (or Cost Explorer), yesterday's charges by service: **not read.** Billing is a console page and the Billing API is also covered by the lock; I have no bill to quote and will not estimate one as if it were real. To fill in tomorrow morning: Billing and Cost Management -> Bills -> charges by service for the day.

Service	USD
Amazon VPC (two interface endpoints)	not read from the bill
Amazon Bedrock	not read from the bill
Lambda, Secrets Manager, CloudWatch	not read from the bill

- Compared with the estimate on Pilot -> Spend: not compared. What I can say from the Terraform files: the two private endpoints cost about USD 0.48 a day whether or not anyone calls the model (a month left running is about USD 15 for nothing); the model itself costs nothing when nobody calls it.

4. What you would do differently for the client's real account

- Run apply and destroy from a pipeline with a plan review, so a destroy always leaves a recorded "Resources: N destroyed" line and the four proof outputs, instead of depending on one person's laptop.
- Keep the Terraform state in an encrypted, versioned bucket with locking, so a teardown can be proved from state and not from "there is no file on my machine". Today the state is local, which is why I cannot show it was ever applied or destroyed.
- Give the checks a read-only role, separate from the one that applies, so verification is not blocked by the same policy that limits the student user.
- Send budget alerts to the bank's FinOps mailbox and have Martin review the budget action before it is switched on.
- Use two Availability Zones and single sign-on, not one shared student user.
- Tag every resource with owner and project (already done in this Terraform) and run a weekly "anything tagged fde-uc1-pilot" sweep, so a forgotten pilot is found by a check, not by a bill.